DECODELABS INDUSTRIAL TRAINING

Project 2: CI/CD pipeline basics

Prepared by:
Onuabueke Martina Chiemezuo

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Introduction

This project focused on learning the fundamentals of Git and GitHub, which are essential tools used in modern software development and DevOps. The project introduced version control concepts, repository management, commit history, and remote collaboration using GitHub. Through a series of practical exercises, I learned how to track changes, manage project history, and synchronize local repositories with remote repositories on GitHub.

Project Objectives

At the end of this project, I was able to:

- Understand what Git is and why version control is important.
- Differentiate between Git and GitHub.
- Configure Git for first-time use.
- Create a local Git repository.
- Track project changes using Git.
- Create commits.
- View repository history.
- Connect a local repository to GitHub.
- Push projects to GitHub using SSH authentication.

What is Git?

Git is a distributed version control system used to track changes in files and source code over time. It enables developers to record project history, collaborate with team members, and restore previous versions when necessary.

Why is Git Important?

Git helps developers:

- Track changes to files.
- Collaborate with other team members.

- Restore previous versions of a project.
- Maintain a complete history of code changes.
- Manage software projects efficiently.

Git vs GitHub

- **Git** is a version control system installed on a local computer.
- **GitHub** is a cloud-based platform used to host and share Git repositories online.

Environment Setup

Operating System:
Ubuntu 24.04.4 LTS running on Windows Subsystem for Linux (WSL)

Tools Used

- Ubuntu Terminal
- Git
- GitHub
- SSH
- Windows 11
- Visual Studio Code (optional if you used it)

Git Version

git --version

Output:

git version 2.43.0

Configure Git

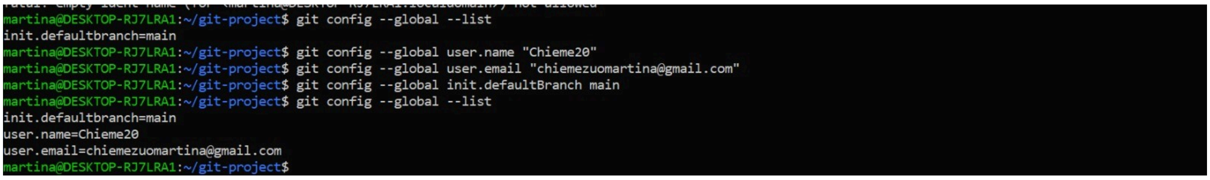
Objective

To configure Git with my username and email address.

Commands

git config --global user.name "Chieme20"git config --global user.email "chiemezuomartina@gmail.com"

git config --global init.defaultBranch maingit config --list



Git requires every commit to contain the author's identity. Configuring the username and email ensures that all commits are correctly attributed.

Real World Application

Before working on projects, developers configure Git so every commit records who made the changes.

Task – Initialize a Git Repository

Objective

To create a new Git repository for tracking changes to a project.

Commands Used

mkdir git-project
cd git-project
git init
ls -la

The `git init` command initializes a new Git repository by creating a hidden `.git` directory. This directory contains all the metadata and version history that Git uses to track changes in the project.

Real-World Application

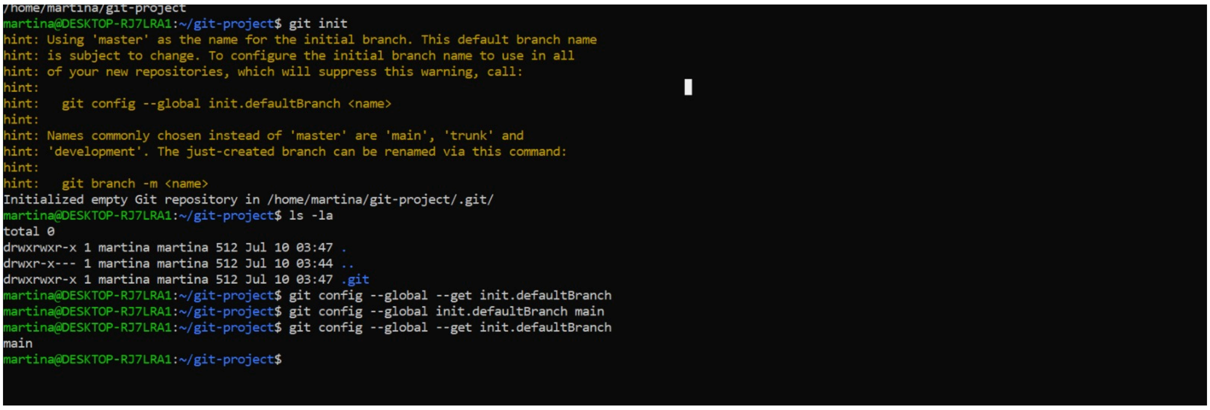
Developers and DevOps engineers use `git init` to begin version-controlling new projects. From that point on, Git can monitor file changes, maintain history, and support collaboration with other team members.

Initialize a Git Repository

To initialize a new Git repository and prepare a project for version control.

Command

git init



The `git init` command initializes a new Git repository by creating a hidden `.git` directory inside the project folder. This directory stores all version history, commits, branches, and configuration required for Git to track changes.

Real-World Application

Developers use `git init` when starting a new project so that every change can be tracked, reviewed, and managed throughout the software development lifecycle.

Staging Files

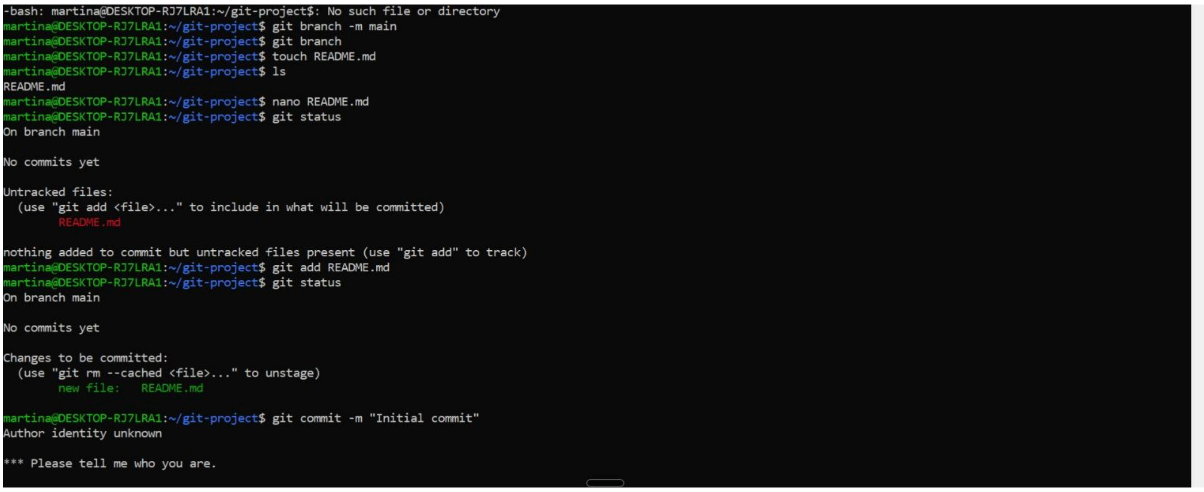
Objective

To prepare files for committing.

Commands

git add README.md

git status



git add moves changes from the working directory into the staging area.

Real World Application

Developers use the staging area to carefully select which changes should be included in the next commit.

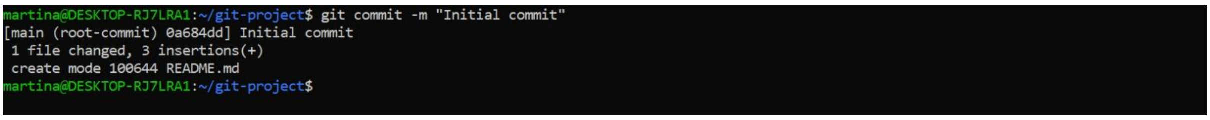
Create the First Commit

Objective

To create the first permanent snapshot of the project by committing the staged changes.

Commands Used

git commit -m "Initial commit"
git status
git log



The `git commit` command saves the staged changes to the Git repository as a permanent snapshot. The `-m` option allows a descriptive commit message to be included. The `git status` command confirms that the working directory is clean, while `git log` displays the repository's commit history.

Real-World Application

Commits enable developers to track the evolution of a project over time. Each commit acts as a checkpoint that can be reviewed, compared, or restored if needed. This is essential for collaboration, debugging, and maintaining software.

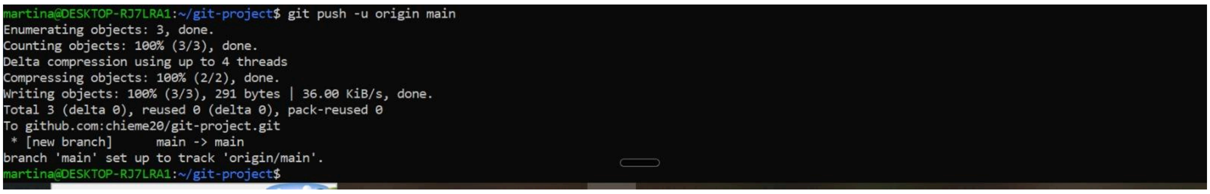
Push the Repository to GitHub

Objective

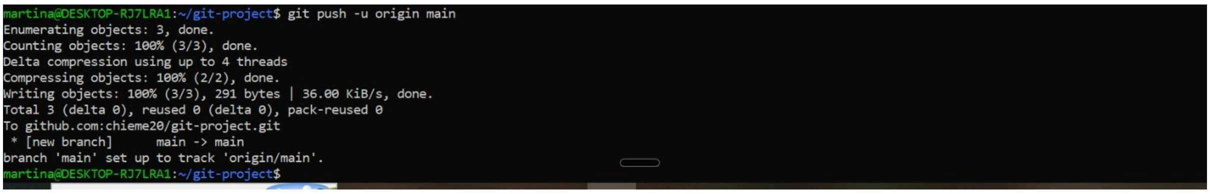
To upload the local Git repository to GitHub using SSH authentication.

Commands Used

ssh -T git@github.com
git push -u origin main



The `ssh -T git@github.com` command tests the SSH connection between the local machine and GitHub. After successful authentication, the `git push -u origin main` command uploads the local repository to GitHub and establishes the `main` branch as the default upstream branch for future pushes.



Real-World Application

SSH authentication provides a secure and convenient way for developers to interact with remote repositories without entering credentials for every push or pull operation. It is widely used in professional software development and DevOps environments.

Checking Repository Status

Objective

To view the current state of the repository.

Command

git status

Explanation

The git status command displays the current state of the repository, including modified files, staged files, and untracked files.

Real World Application

Developers run git status before almost every commit to understand what has changed.

Viewing Differences

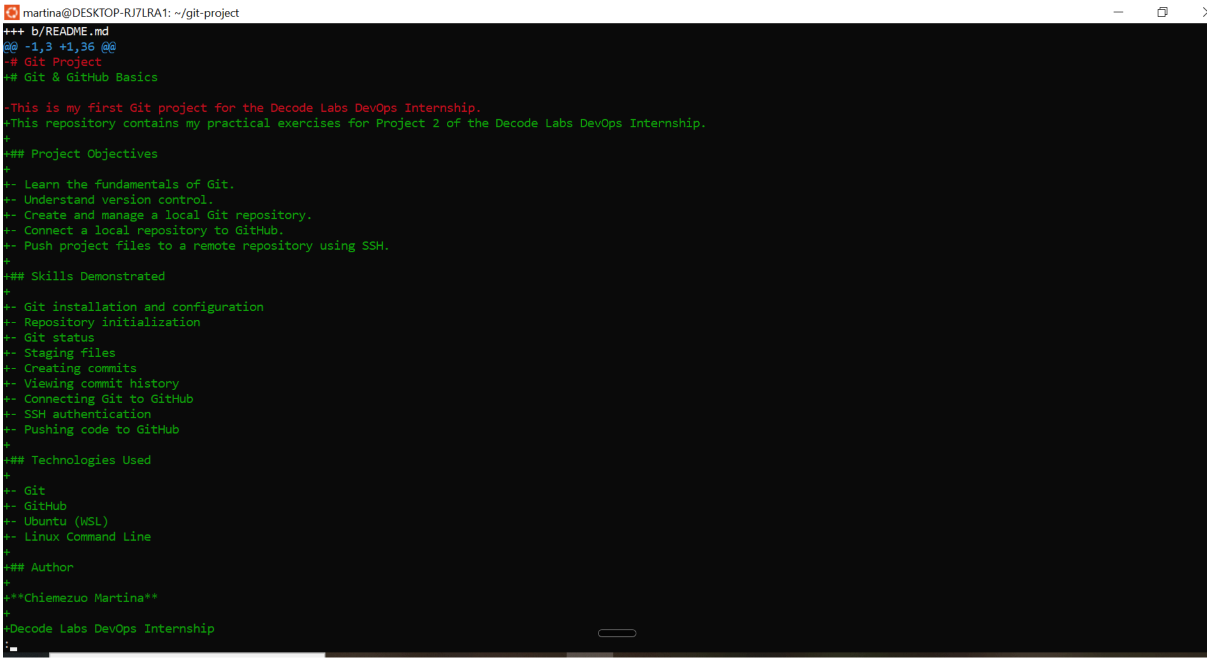
Objective

To compare changes made to files before committing.

Command

git diff

git diff compares the working directory with the most recent commit and displays all modifications.


martina@DESKTOP-RJ7LRA1: ~/git-project
+++ b/README.md
@@ -1,3 +1,36 @@
-# Git Project
+# Git & GitHub Basics

-This is my first Git project for the Decode Labs DevOps Internship.
+This repository contains my practical exercises for Project 2 of the Decode Labs DevOps Internship.
+
+## Project Objectives
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+- Learn the fundamentals of Git.
+- Understand version control.
+- Create and manage a local Git repository.
+- Connect a local repository to GitHub.
+- Push project files to a remote repository using SSH.
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+## Skills Demonstrated
+
+- Git installation and configuration
+- Repository initialization
+- Git status
+- Staging files
+- Creating commits
+- Viewing commit history
+- Connecting Git to GitHub
+- SSH authentication
+- Pushing code to GitHub
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+## Technologies Used
+
+- Git
+- GitHub
+- Ubuntu (WSL)
+- Linux Command Line
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+## Author
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+**Chiemezuo Martina**
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+Decode Labs DevOps Internship
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